



AKTU

NAS



Unit -I



MOST IMPORTANT QUESTIONS

III-Sem (BEC303 - NETWORK ANALYSIS AND SYNTHESIS)

AKTU : Syllabus

Unit-I

Node and mesh analysis, matrix approach of network containing voltage & current sources and reactances, source transformation and duality.

Q.1 Calculate V_o across 4 ohm resistance.

apply KCL at Node A

$$4I_o = 10 \times 10^{-3} + I + I_o$$

$$3I_o = 10 \times 10^{-3} + \frac{V_A - 0}{(2+4) \times 10^3}$$

$$3 \left[\frac{V_A - 0}{3 \times 10^3} \right] = 10 \times 10^{-3} + \frac{V_A}{6 \times 10^3}$$

$$V_A \times 10^{-3} = 10^{-3} \left[10 + \frac{V_A}{6} \right]$$

$$V_A - \frac{V_A}{6} = 10$$

$$\frac{6V_A - V_A}{6} = 10 \Rightarrow 5V_A = 60$$

$$V_A = \frac{60}{5} = 12 \text{ Volt}$$

$$I = \frac{V_A}{6 \times 10^3} = \frac{12}{6 \times 10^3}$$

$$I = 2 \text{ mA}$$

V_o = Voltage across 4k Ω

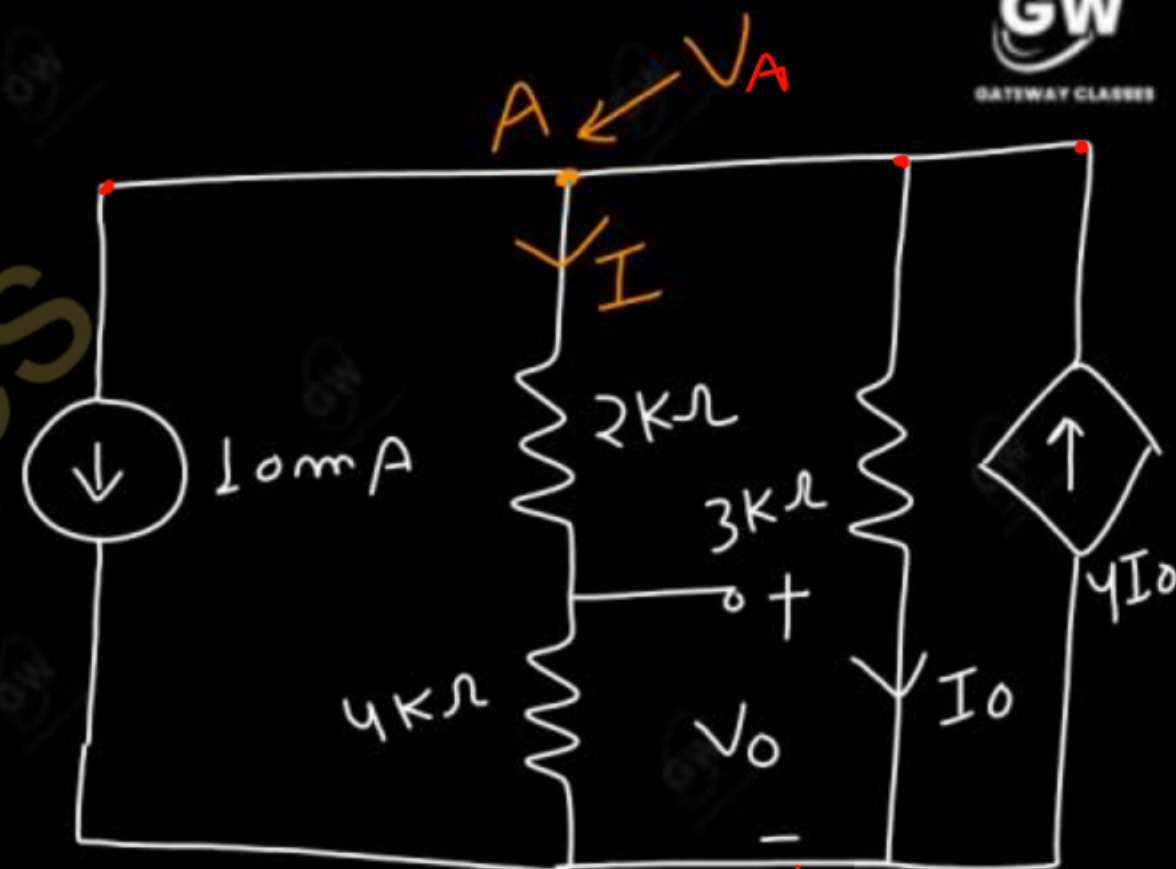
from ohm's Law

$$V_o = I \times 4 \times 10^3$$

$$= 2 \times 10^{-3} \times 4 \times 10^3$$

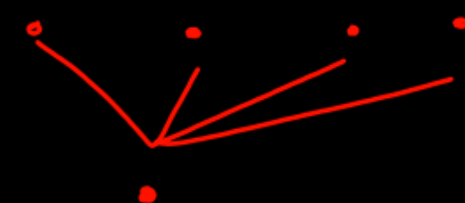
$$V_o = 8 \text{ Volt}$$

Ans



$V = IR$ ✓

R.f. Node $V_o = 0 \text{ V}$



Q.2 Identify the node voltages in the circuit shown in the fig..

Step-1

According to KCL at Node 1

$$3 = i_x + I_1$$

$$3 = \frac{V_1 - V_2}{2} + \frac{V_1 - V_3}{4}$$

$$3 = \frac{2V_1 - 2V_2 + V_1 - V_3}{4}$$

$$3V_1 - 2V_2 - V_3 = 12 \quad \text{--- (1)}$$

Step-2

According to KCL at Node 2

$$i_x = I_2 + I_3$$

$$\frac{V_1 - V_2}{2} = \frac{V_2 - 0}{4} + \frac{V_2 - V_3}{8}$$

$$\frac{V_1 - V_2}{2} = \frac{2V_2 + V_2 - V_3}{8}$$

$$4V_1 - 4V_2 = 3V_2 - V_3$$

$$4V_1 - 7V_2 + V_3 = 0 \quad \text{--- (2)}$$

Step-3, Apply KCL at (3)

$$I_1 + I_3 = 2i_x$$

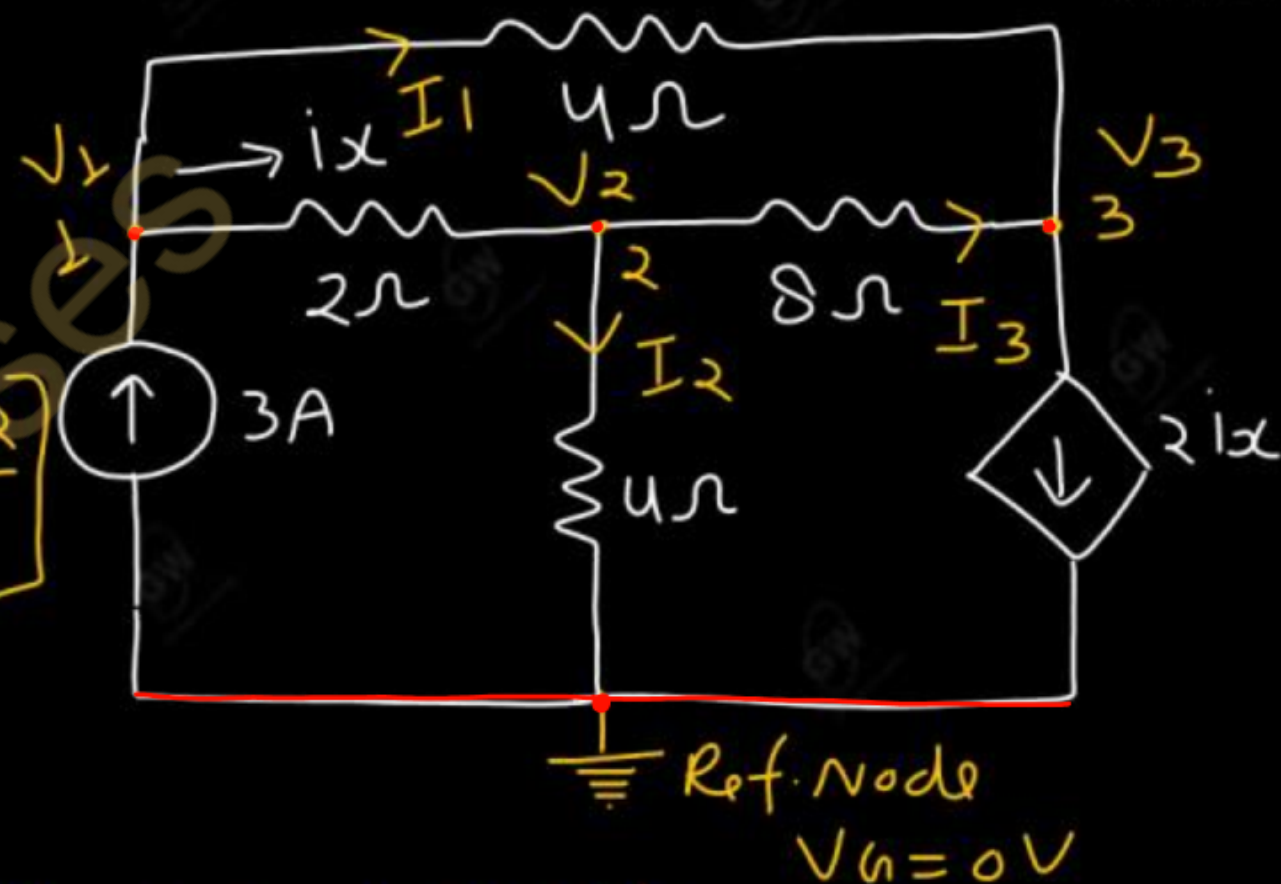
$$\frac{V_1 - V_3}{4} + \frac{V_2 - V_3}{8} = 2 \left[\frac{V_1 - V_2}{2} \right]$$

$$\frac{2V_1 - 2V_3 + V_2 - V_3}{8} = V_1 - V_2$$

$$2V_1 + V_2 - 3V_3 - 8V_1 + 8V_2 = 0$$

$$-6V_1 + 9V_2 - 3V_3 = 0 \quad \text{--- (3)}$$

after solving (1), (2) and (3)



$$V_1 = 4.8 \text{ Volt}$$

$$V_2 = 2.4 \text{ Volt}$$

$$V_3 = -2.4 \text{ Volt}$$

Ans
↓
Node Voltages

Q.3 Calculate the mesh current i_1 and i_2 in the circuit as shown in fig. GW

Step-1

According to KVL in
mesh ① [A B C D A]

$$\underline{-12 + 2i_1 + 12(i_1 - i_2)} \\ \underline{+ 4i_1 = 0}$$

$$18i_1 - 12i_2 = 12 \quad \text{--- (1)}$$

Step-2

According to KVL in mesh ②

$\Rightarrow$ mesh ② [D C E F D]

$$\underline{-12(i_1 - i_2) + 9i_2 + 8} \\ \underline{+ 3i_2 = 0}$$

$$\underline{-12i_1 + 12i_2 + 9i_2} \\ \underline{+ 8 + 3i_2 = 0}$$

$$\underline{-12i_1 + 24i_2 = -8}$$

after solving

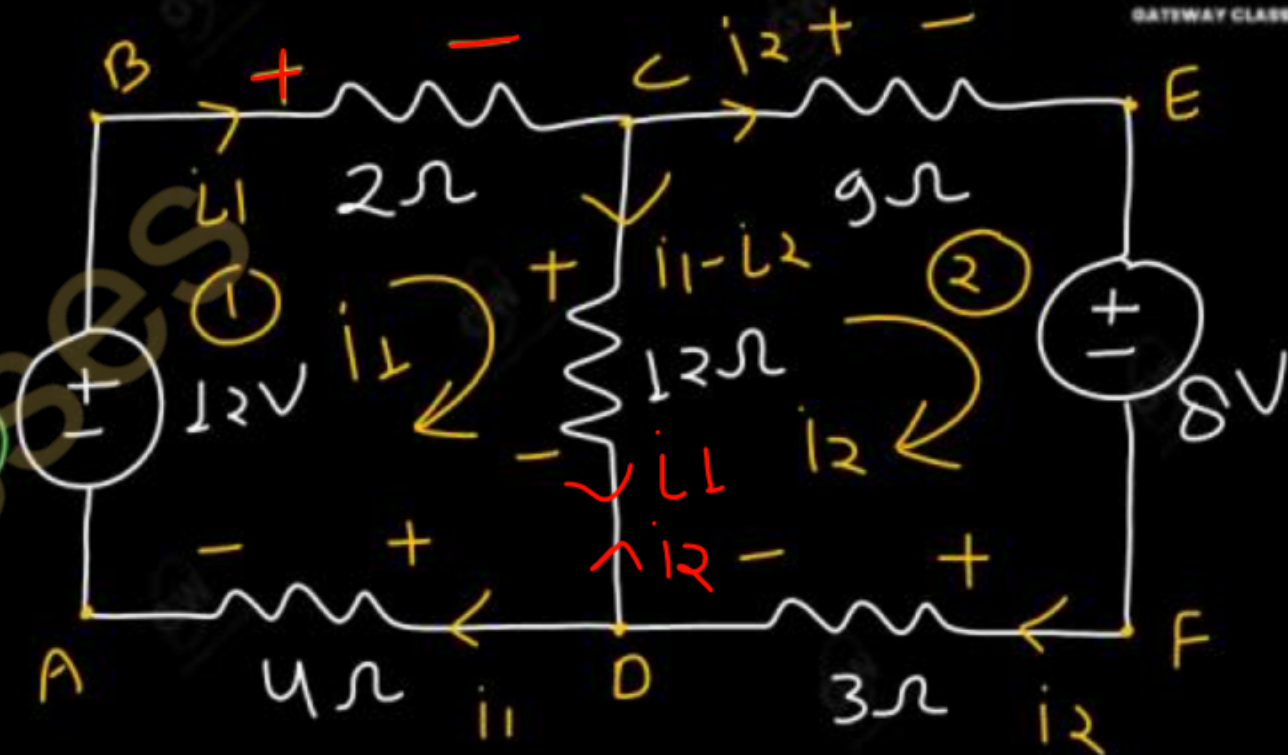
① and ②, we have

$$\checkmark \underline{i_1 = 0.666 \text{ Amp}}$$

$$\checkmark \underline{i_2 = 0 \text{ A}}$$

Ans

mesh currents



$i_1 \rightarrow$ current in mesh ①

$i_2 \rightarrow$ current in mesh ②

Q.4 Identify the node voltages in the circuit shown in the fig..

Step-1

According to KCL at Node 1

$$S = I_1 + I_2$$

$$S = \frac{V_1 - 0}{2} + \frac{V_1 - V_2}{4}$$

$$S = \frac{2V_1 + V_1 - V_2}{4}$$

$$3V_1 - V_2 = 20 \quad (1)$$

Step-2

apply KCL at Node 2

$$I_2 + 10 = S + I_3$$

$$I_2 - I_3 = -5$$

$$\frac{V_1 - V_2}{4} - \left(\frac{V_2 - 0}{6} \right) = -5$$

$$\frac{3V_1 - 3V_2 - 2V_2}{12} = -5$$

$$3V_1 - 5V_2 = -60 \quad (2)$$

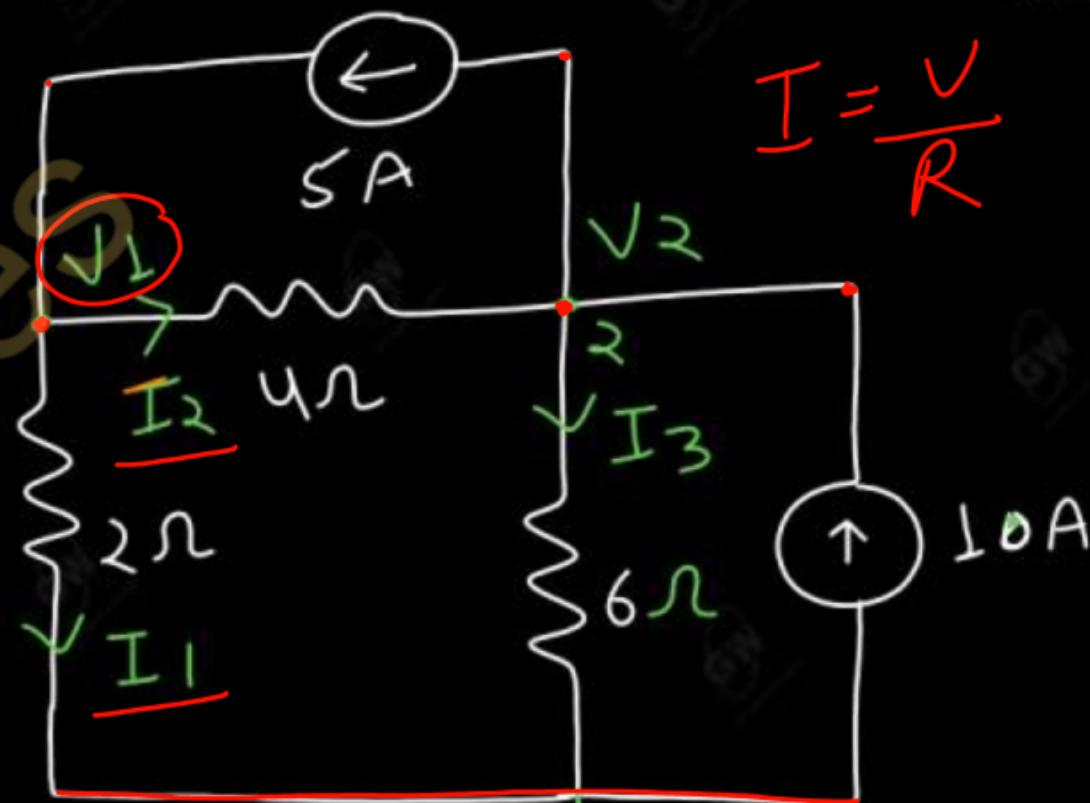
after solving eq(n) (1) and (2)

$$V_1 = 13.33 \text{ Volt}$$

$$V_2 = 20 \text{ Volt}$$

Ans

↓
Node Voltages



$$I = \frac{V}{R}$$

Ref. Node
 $V_0 = 0V$

V_1 = Voltage at Node 1
 V_2 = Voltage at Node 2

Q.5 For the circuit given in the fig, find the branch currents I_1, I_2 and I_3 using mesh analysis.

Step-1

Apply KVL in mesh (1) [ABCD A]

$$-15 + 5I_1 + 10(I_1 - I_2) + 10 = 0$$

$$15I_1 - 10I_2 = 5 \quad (1)$$

Step-2

Apply KVL in mesh (2) [DCEFD]

$$-10 - 10(I_1 - I_2) + 6I_2 + 4I_2 = 0$$

$$-10I_1 + 20I_2 = 10 \quad (2)$$

after solving eqn (1) and eqn (2)

we have

$$I_1 = 1A$$

$$I_2 = 1A$$

mesh currents

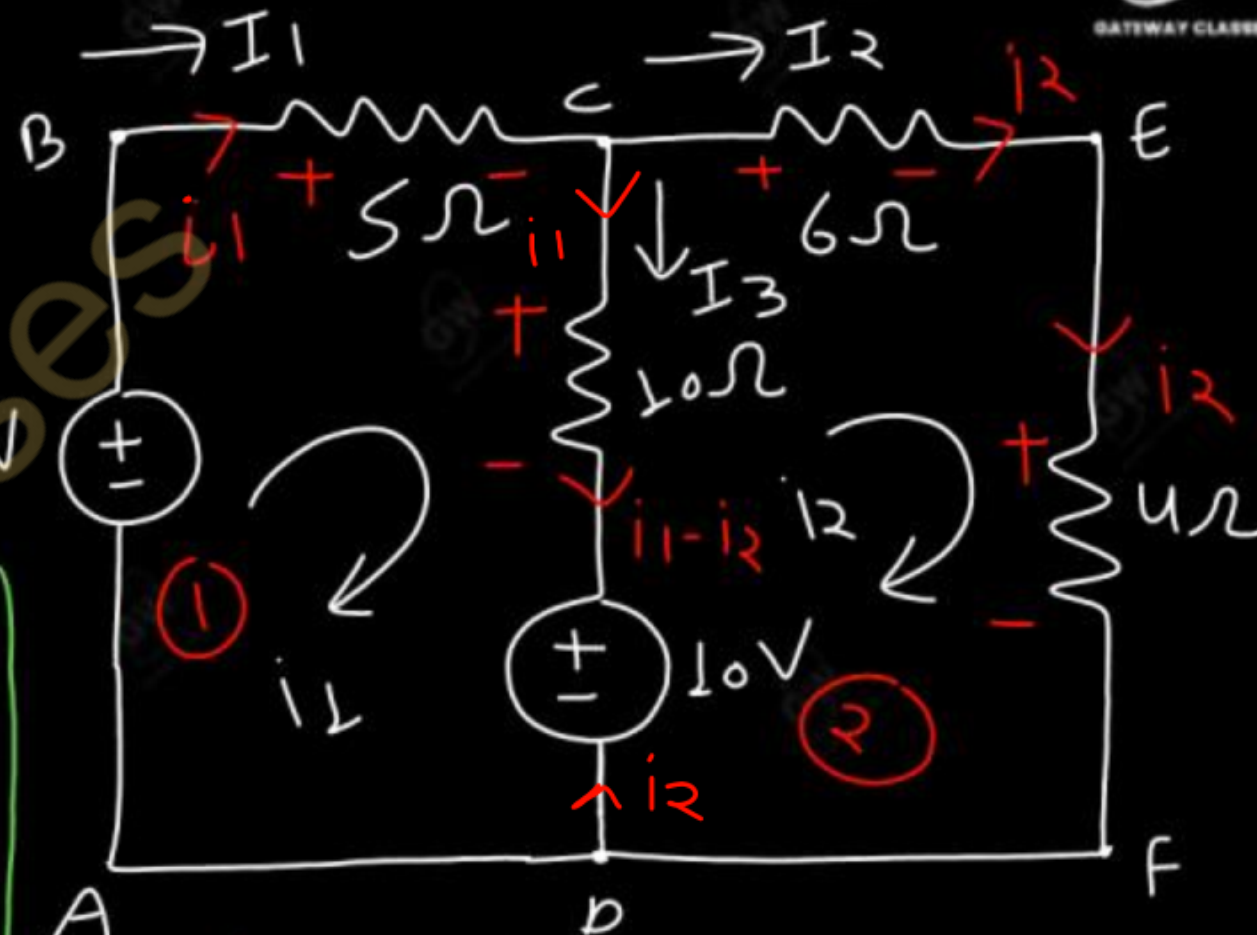
$$I_1 = I_1 = 1A$$

$$I_2 = I_2 = 1A$$

$$I_3 = I_1 - I_2 = 0A$$

Ans

- ✓ $I_1 = \text{current in } 5\Omega$
- ✓ $I_2 = \text{current in } 6\Omega$
- ✓ $I_3 = \text{current in Branch CD}$

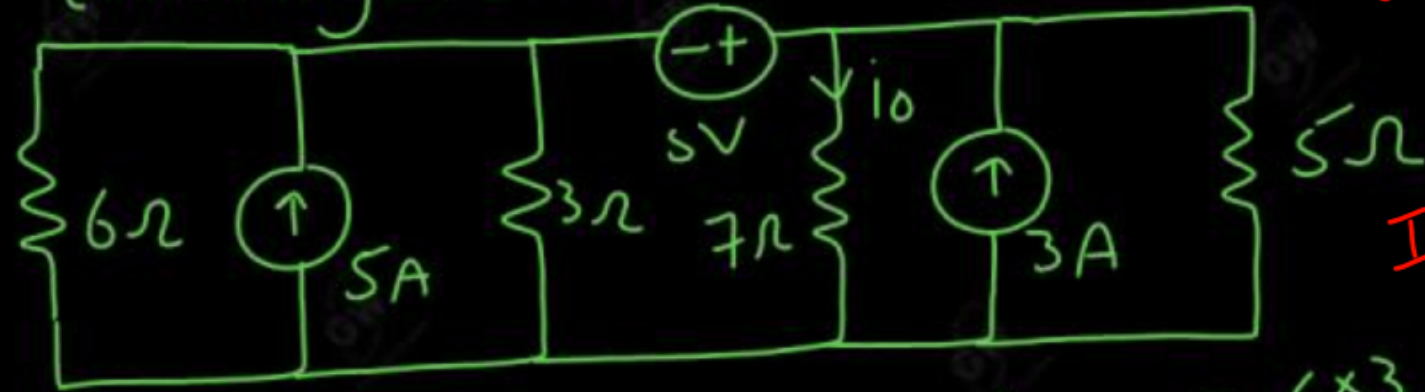


$I_1 = \text{current in mesh (1)}$
 $I_2 = \text{current in mesh (2)}$

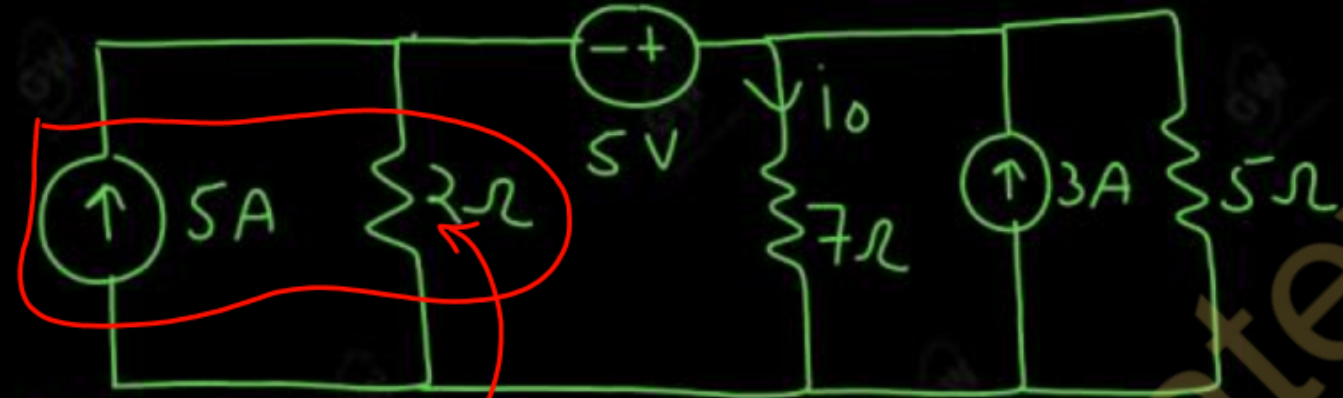
Q.6 Define Source Transformation and calculate I_0 in the circuit using Source Transformation

Step-1

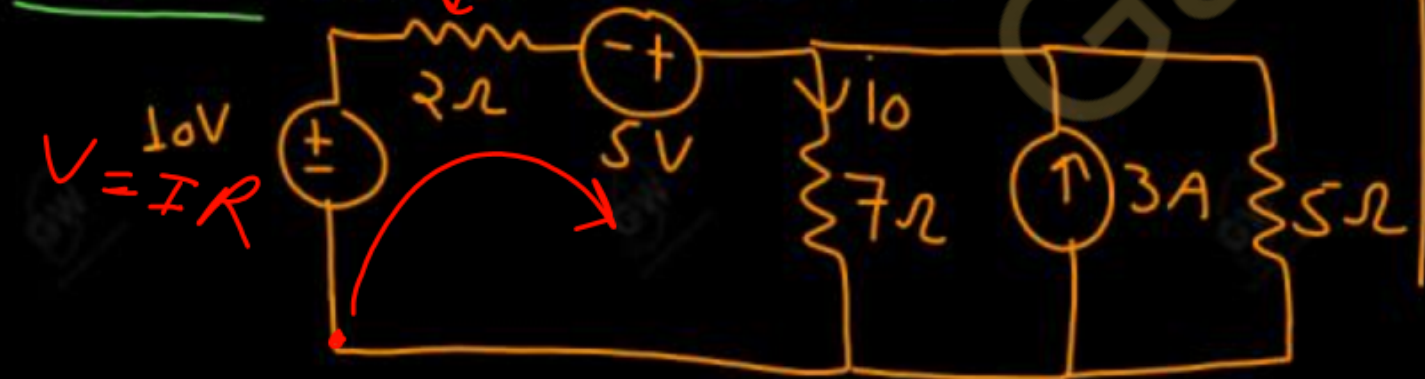
combining 1Ω and $4\Omega \Rightarrow 1+4=5\Omega$



Step-2 combining 6Ω and $3\Omega \Rightarrow \frac{6 \times 3}{9} = 2\Omega$

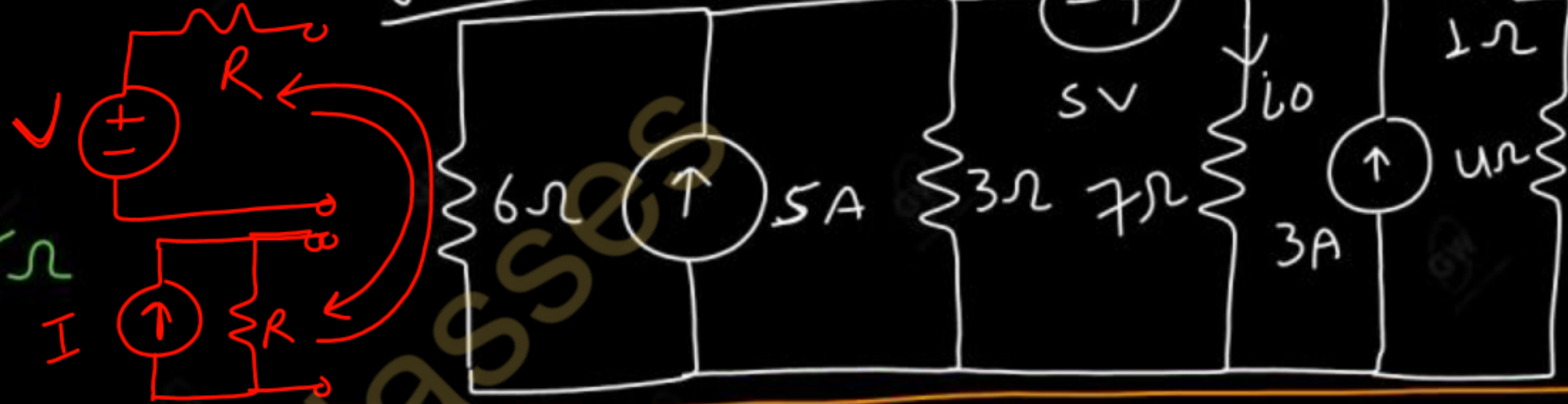


Step-3 convert 5A CS into VS



$$V = IR$$

Given circuit



Step-4

combining 10V and 5V



Step-5 Transform 15V VS



$$I = \frac{V}{R}$$

Step-6



from current div Rule

$$I_0 = 7.5 \times \left(\frac{10}{7} \right)$$

$$= \frac{15}{\frac{10}{7} + 7} = 1.779 \text{ Amp}$$

Explain the principle of Duality and Draw the Dual circuit for the given network.

Sol

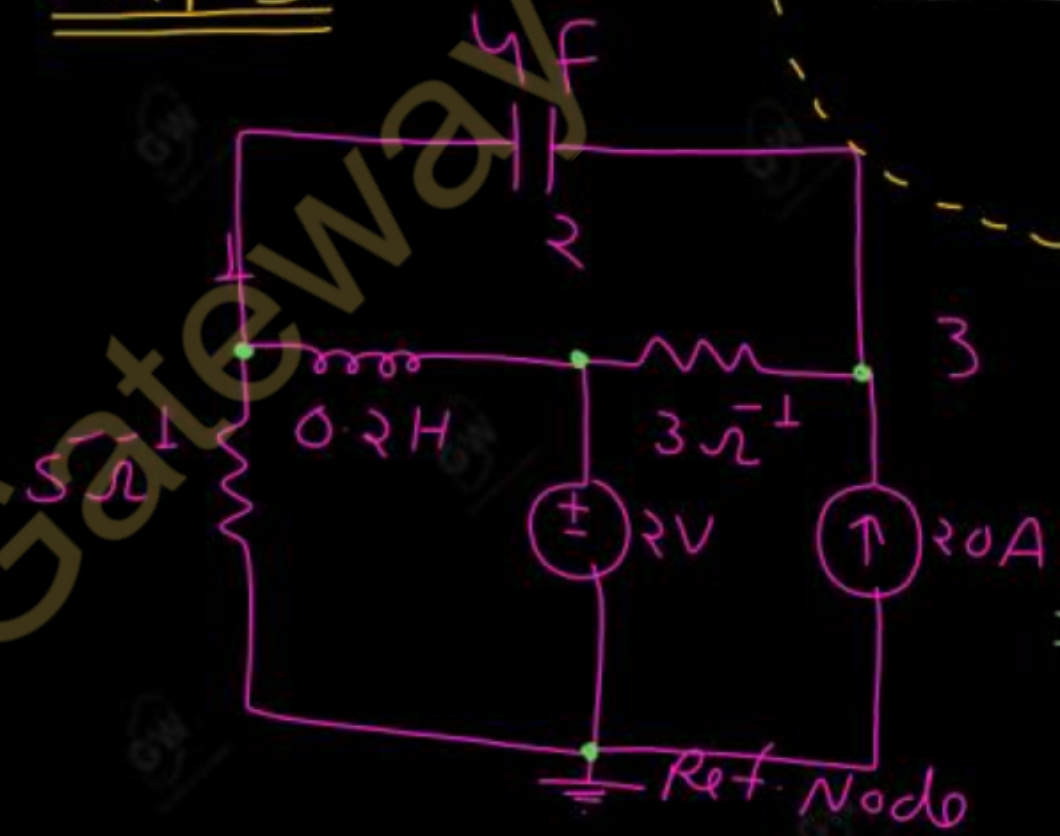
Duality Table

Step-1

Elements	Dual elements
R	G Conductance
L	C
C	L
V	I
I	V
Series	Parallel
Node	mesh
Thermin's	Norton

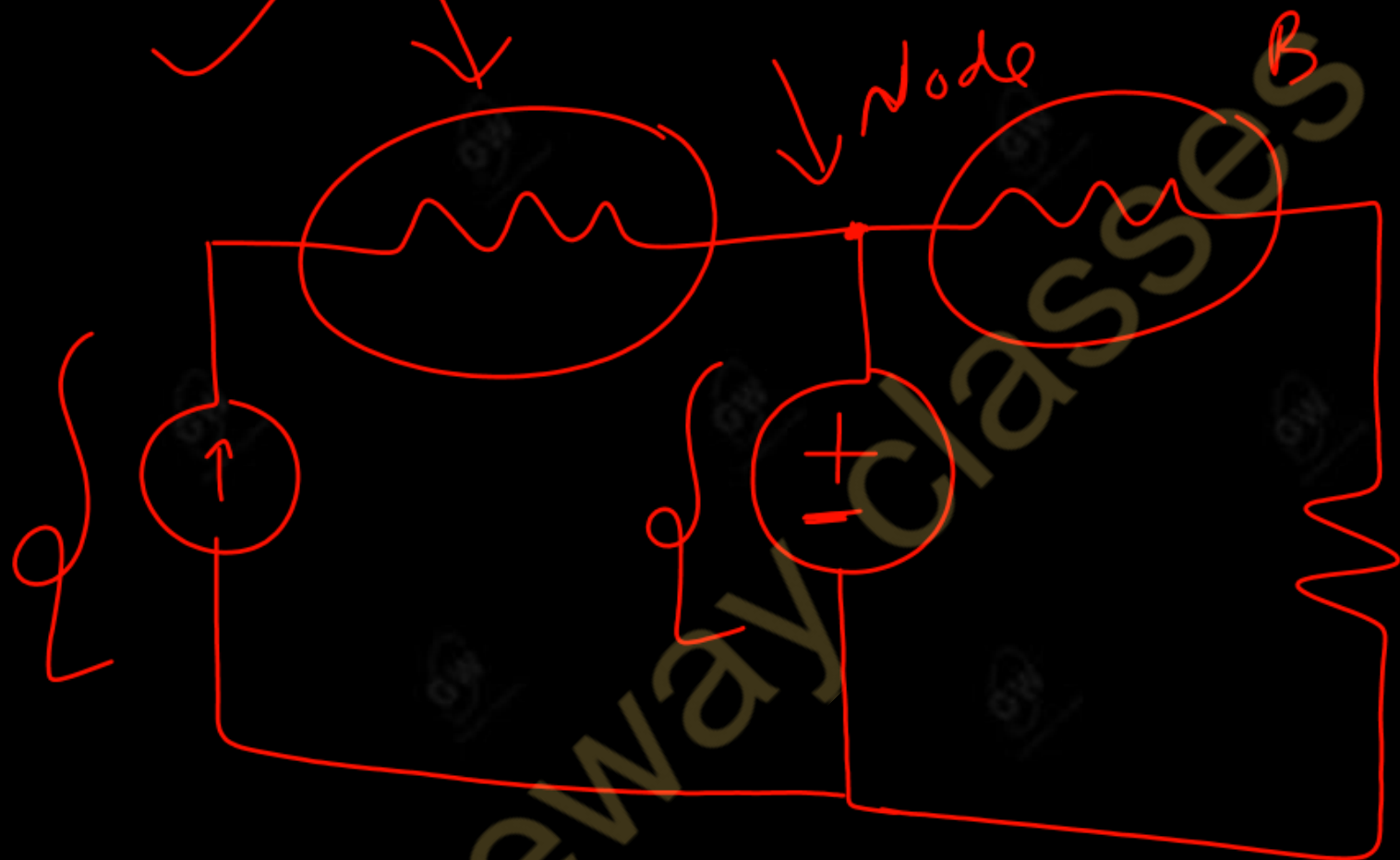
Step-3

Step-2



⇒ final dual ckt for given network

Q.8 Define node, Branch, Active and Passive network, linear and non-linear network..



AKTU Full Courses (Paid)
Download **Gateway Classes** Application
From Google Play store
All Subjects

Link in Description

Thank You